

Institute of Actuaries of India

Subject CT4 – Models

May 2013 Examinations

INDICATIVE SOLUTIONS

Solution 1 :-

- i. In assessing the suitability of a model for a particular exercise it is important to consider the following:
- The objectives of the modelling exercise.
 - The validity of the model for the purpose to which it is to be put.
 - The validity of the data to be used.
 - The possible errors associated with the model or parameters used not being a perfect representation of the real world situation being modelled.
 - The impact of correlations between the random variables that “drive” the model.
 - The extent of correlations between the various results produced from the model.
 - The current relevance of models written and used in the past.
 - The credibility of the data input.
 - The credibility of the results output.
 - The dangers of spurious accuracy.
 - The ease with which the model and its results can be communicated.
- ii. There are two approaches to the task of simulating a **time-homogeneous** Markov jump process. The first is an approximate method and the second exact.

Approximate method

Divide time into very short intervals of width h , say, where $\mu_{ij}h$ is much smaller than 1 for each i and j . The transition matrix $P(h)$ of the Markov chain has entries approximately given by:

$$p_{ij}^*(h) = \delta_{ij} + \mu_{ij}h$$

We may simulate a discrete-time Markov chain $(Y_n, n \geq 0)$ with these transition probabilities, and then write $X_t = Y_{\lfloor \frac{t}{h} \rfloor}$

This simplistic method is not very satisfactory, as its long-term distribution may differ significantly from that of the process being modelled.

Exact method

This takes advantage of the structural decomposition of the jump process. First simulate the jump chain of the process as a Markov chain with transition probabilities $p_{ij}(h) = \frac{\mu_{ij}}{\lambda_i}$. Once the path $\{X_n: n = 0, 1, \dots\}$ has been generated, the holding times $\{T_n: n = 0, 1, \dots\}$ are a sequence of independent exponential random variables, T_n having rate parameter given by λ_{X_n}

[Total 6 Marks]

Solution 2 :-**i. Importance of dividing the data into homogeneous classes**

Mortality investigations aim to provide insight in to the expected probabilities of death and survival for a given population. This study is enriched by considering the underlying causes of mortality. As a reasonable proxy for studying the causes, we can choose various risk drivers such as age, gender, smoker status, data for which are relatively easier to gather. Dividing the data into such homogenous classes helps identify various trends within the mortality rates driven by the risk drivers and can thus help identify and manage specific risks within the overall portfolio. In particular, this helps in identifying appropriate risk-based weightings at a more granular level than otherwise possible if heterogeneous data is used.

ii. Comment on European gender directive

Construction of a single table for both males and females has the advantage of relative simplicity within the basis as well as provide a direct way of compliance with the European ruling on gender. However, any such “combined table” would necessarily rely on certain assumptions about the underlying business mix in respect of gender distribution within the insurer’s portfolio. It is possible that over time, this mix changes and therefore the use of the combined table based on different weightings may no longer be appropriate. Moreover, statistically, mortality rates between males and females have been demonstrably different. Therefore, whilst a combined table may well be suitable for the purpose of pricing of new contracts, it may not be so for other actuarial investigations, such as profit testing, reserving, risk management and setting capital requirements, other valuations etc. In such investigations, it may still be desirable to explicitly identify the risk factors presented by individual sex categories and make an allowance for them separately. In addition, it may be of interest for the actuary to monitor the on-going risk from changing business mix as well as any possible selection risks – for which data, identifiable by gender would be necessary.

Therefore, we could only partly agree with the Appointed Actuary’s suggestion to construct a combined mortality table – whilst arguably this would be useful for the purpose of compliance, it may still be worthwhile having more granular risk-factor based rates for other actuarial investigations, particularly those in respect of risk and capital management.

[Total 6 Marks]**Solution 3 :-****i. Expression for ${}_tq_x$**

- Uniform distribution of deaths: ${}_tq_x = t \cdot q_x$
- Constant force of mortality: ${}_tq_x = 1 - e^{-\mu t}$
- Balducci assumption: ${}_tq_x = \frac{t \cdot q_x}{1 - (1-t) \cdot q_x}$

ii. Probability of death in the first three months of birth, ${}_{0.25}q_0$

From the information provided, $q_0 = 0.067$

a. Under uniform distribution of deaths, ${}_{0.25}q_0 = 0.25 \times 0.067 = 0.01675$

b. Under constant force of mortality,

$$\mu = -\log(1 - 0.067) \Rightarrow {}_{0.25}q_0 = 1 - (1 - 0.067)^{0.25} = 0.01719$$

c. Under Balducci assumption, ${}_{0.25}q_0 = \frac{0.25 \times 0.067}{1 - (1 - 0.25) \times 0.067} = 0.01764$

iii. Appropriateness of assumption for infant mortality

The Balducci assumption implies that the force of mortality is decreasing over the first year of birth, which means that the mortality is heavier in the first half of the year than in the second half. This is probably true for infant mortality as the mortality should be expected to reduce once the critical initial few months have passed, i.e. mortality should be heaviest immediately (or soon after birth) and decrease over time. Therefore, Balducci assumption should be considered most appropriate for a model of infant mortality.

[Total 6 Marks]

Solution 4 :-

i. complete expectation of life, e_x

$$e_x = E[T_x] = \int_0^{\omega-x} {}_t p_x dt$$

This represents the integral of the probability of survival at each future age, i.e. the expected future lifetime of a life currently aged x . In other words, this is the expectation of life at age x or how many years a life is expected to live given that it is currently x years old.

ii. The curtate expectation of life

$$e_x = \sum_{k=1}^{\infty} {}_k p_0 = \sum_{k=1}^{\infty} e^{-0.0325k} = \frac{e^{-0.0325}}{1 - e^{-0.0325}} = 30.27$$

iii. The probability that a life aged exactly 36 will survive to age 45.

$${}_9 p_{36} = \exp \left[- \int_0^9 0.0325 dt \right] = e^{-0.2925} = 0.7464 \approx 75\%$$

iv. The exact age x representing the median of the life-time T of a new born baby.

The median of the life-time T implies that the probability, ${}_x p_0 = 0.5$

$$\text{Thus, } {}_x p_0 = 0.5 \Rightarrow \exp(x \cdot -0.0325) = 0.5 \Rightarrow x = -\frac{\log 0.5}{0.0325} = 21.33$$

[Total 8 Marks]

Solution 5 :-

The transition probability matrix is given as

$$\begin{pmatrix} .75 & .25 \\ .2 & .8 \end{pmatrix}$$

Also, we have $X(0) = 2$ (inactive mode) with probability 1, i.e. $P_0 = (0,1)$

$$i. \quad P_2 = P_0 P^2 = (0,1) \begin{pmatrix} .75 & .25 \\ .2 & .8 \end{pmatrix}^2$$

- ii. To find the steady-state distribution, we solve the system of equations

$$\begin{aligned}\pi P &= \pi \\ \pi_1 + \pi_2 &= 1\end{aligned}$$

These equations are

$$0.75\pi_1 + 0.2\pi_2 = \pi_1$$

$$0.25\pi_1 + 0.8\pi_2 = \pi_2$$

$$\pi_1 + \pi_2 = 1$$

One of the equations is redundant.

From the first equation,

$$\begin{aligned}0.25\pi_1 &= 0.2\pi_2 \\ \pi_1 &= 0.8\pi_2\end{aligned}$$

Substituting in the third equation,

$$0.8\pi_2 + \pi_2 = 1$$

$$\pi_2 = 0.556$$

Hence,

$$\begin{aligned}\pi_1 &= 0.555 \times .8 \\ \pi_1 &= 0.444\end{aligned}$$

In a steady state,

$$P(X_n = \text{Active}) = 0.444$$

$$P(X_n = \text{Inactive}) = 0.556$$

[Total 6 marks]

Solution 6 :-

- i. Let C be the number of Type I defects for a lot of 20 semiconductors.

C has a Poisson distribution with mean rate = 3/100 (Type I defects per semiconductor).

The packet contains 20 semiconductors.

Hence, the parameter $\lambda = (3/100)(20) = 0.6$,

Thus

$$\begin{aligned}P(C = 2) &= e^{-0.6} \left(\frac{0.6^2}{2!} \right) \\ &= 0.099\end{aligned}$$

- ii. Let S denote the number Type II defects for a lot of 20 semiconductors.

C has a Poisson distribution with mean rate = $5/100$ (Type II defects per semiconductor).

The packet contains 20 semiconductors.

Hence the parameter $\lambda = (5/100)(20) = 1$,

Thus the probability of finding one or more defects

$$\begin{aligned} 1 - P(S = 0) &= 1 - e^{-1} \\ &= 0.632 \end{aligned}$$

- iii. Let X is the total number of defects for pack of 20 semiconductors.

Hence the mean rate of defects would be = $(3 + 5)/100$ or = 0.08 defects per semiconductor,

This is multiplied to the number of semiconductor in a pack of 20.

Hence the parameter $\lambda = 1.6$

$$\begin{aligned} P(X > 2) &= 1 - P(X \leq 2) \\ &= 1 - e^{-1.6} \left(1 + 1.6 + \frac{1.6^2}{2!} \right) \\ &= 1 - 0.7834 = 0.2166 \end{aligned}$$

- iv. Let N be the total number of packs rejected out of 4 packets.
 N should follow a binomial distribution
 with $n = 4, p = \text{probability of one packet getting rejected}$.

Thus $p = 0.2166$

$$\begin{aligned} P(N = 1) &= \binom{4}{1} \times 0.2166 \times (1 - 0.2166)^3 \\ &= 0.4166 \end{aligned}$$

[Total 10 marks]

Solution 7 :-

- i. If $i = 1$ and it's *raining*, then the shopkeeper takes his umbrella and moves to the other place. When he reaches the other place (shop or home as the case may be), he will end up with three umbrellas at that place.

If p is the probability of *raining* then the shopkeeper will carry the umbrella with probability p if there are 1 or 2 umbrellas from where the shopkeeper is presently starting.

If there are three umbrellas from where the shopkeeper is presently starting, then the probability of carrying umbrella is $p' = 0.25(1 + 3p)$ and probability of not carrying an umbrella is $0.75(1 - p)$, then.

$P_{i,j}$ is the probability of carrying umbrella from state i to state j .

Thus,

$$P_{1,3} = p$$

If $i = 1$ and it's not a *raining* and shopkeeper is not carrying an umbrella, he goes to the other place and find 2 umbrellas. Thus,

$$P_{1,2} = q$$

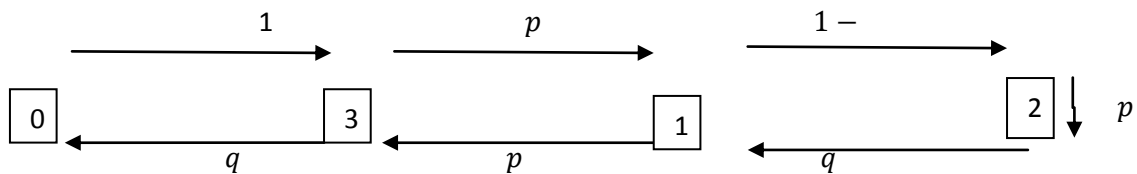
where $q = 1 - p$

And starting from 3 umbrellas,

$$P_{3,1} = p' = 0.25(1 + 3p)$$

$$P_{3,0} = q' = 0.75(1 - p)$$

Continuing in the same manner I form a Markov chain with the following diagram:



The transition matrix is given by :

$$\begin{array}{c}
 \begin{matrix} & 0 & 1 & 2 & 3 \end{matrix} \\
 \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & q & p \\ 0 & q & p & 0 \\ q' & p' & 0 & 0 \end{pmatrix}
 \end{array}$$

ii. In order to find stationary distribution we solve the system of equations

$$\begin{aligned}
 \pi P &= \pi \\
 \pi_0 + \pi_1 + \pi_2 + \pi_3 &= 1
 \end{aligned}$$

$$\pi_0 = q' \pi_3$$

$$\pi_3 = \pi_0 + p\pi_1$$

$$\pi_1 = q \pi_2 + p' \pi_3$$

$$\pi_2 = q \pi_1 + p\pi_2$$

Solving the last equation,

We get $\pi_1 = \pi_2$

Also,

$$\pi_1 = \frac{.25(1+3p)\pi_3}{p} = \pi_2$$

Using this in the $\sum \pi$ we get,

$$2\pi_1 + \pi_3 + q'\pi_3 = 1$$

$$2 \frac{.25(1+3p)\pi_3}{p} + \pi_3 + 0.75q\pi_3 = 1$$

$$\pi_3 = \frac{p}{2 \times .25(1+3p) + p + 0.75pq}$$

$$\pi_0 = \frac{0.75(1-p)}{2 \times .25(1+3p) + p + 0.75pq}$$

Every time the shopkeeper gets wet when he happens to be in state 0 and it's *raining*. The probability that he is in state 0 is π_0 and the probability that it's raining, is p .

Hence, the required probability is

$$p \cdot \pi_0 = \frac{0.75(1-p)p}{2 \times .25(1+3p) + p + 0.75pq}$$

iii.

Now we know that, $p = 0.7$

Hence, the required probability is 4.58%

If I want the chance to be less than 2% then, clearly, I need more umbrellas.

So, suppose he needs N umbrellas. By following the same Markov chain (as above) for N umbrellas, we find that

$$\pi_{N-1} = \pi_{N-2} \dots \dots \dots = \pi_2 = \pi_1$$

Also

$$\pi_1 = \frac{0.25(1+3p)\pi_N}{p} = \pi_2 \dots \dots = \pi_{N-1}$$

And

$$\pi_0 = q' \pi_N$$

Using this in the $\sum \pi$ we get,

$$(N-1) \pi_1 + \pi_N + q' \pi_N = 1$$

$$(N-1) \frac{0.25(1+3p)}{p} \pi_N + \pi_N + q' \pi_N = 1$$

$$(N-1)0.25(1+3p) \pi_N + p\pi_N + pq'\pi_N = p$$

$$\pi_N = \frac{p}{(N-1) \times .25(1+3p) + p + 0.75pq}$$

Hence,

$$\pi_0 = \frac{0.75p(1-p)}{(N-1) \times .25(1+3p) + p + 0.75pq}$$

The Probability that the shopkeeper gets wet $\pi_0 p$, which is

$$\frac{0.75(1-p)p^2}{(N-1) \times .25(1+3p) + p + 0.75pq}$$

As we want the required probability to be less 2%,

$$0.75(1-p)p^2 < 2\% \{ (N-1) \times .25(1+3p) + p + 0.75pq \}$$

$$75(1-p)p^2 < 2 \{ (N-1) \times .25(1+3p) + p + 0.75pq \}$$

$$75(1-p)p^2 < 0.5(N-1)(1+3p) + 2p + 1.5pq$$

$$75(1-p)p^2 - 2p - 1.5pq < 0.5(N-1)(1+3p)$$

$$1 + \frac{75(1-p)p^2 - 2p - 1.5pq}{0.5(1+3p)} < N$$

Solving the above equation gives $N=7$.

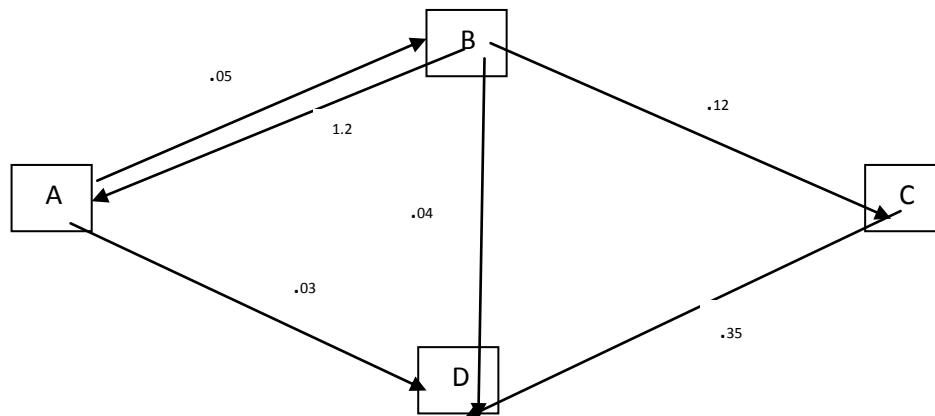
So to reduce the chance of getting wet to less than 2% the shopkeeper needs 7 umbrellas.

[Total 15 marks]

Solution 8 :-

i.

$$\begin{array}{c}
 \begin{matrix} & A & B & C & D \end{matrix} \\
 \begin{matrix} A \\ B \\ C \\ D \end{matrix} \begin{pmatrix} -.08 & .05 & 0 & .03 \\ 1.2 & -1.36 & .12 & .04 \\ 0 & 0 & -.35 & .35 \\ 0 & 0 & 0 & 0 \end{pmatrix}
 \end{array}$$



- ii. The required event is that either no jump out of A has taken place by time 4 or that a jump to D has taken place.

The probability that no jump had taken place out of A by 4 years' time

$$e^{-0.08 \times 4}$$

$$= 0.7261$$

The probability that a jump had taken place out of A only to D by time 4

$$(1 - e^{-0.08 \times 4}) \times \frac{0.3}{0.8}$$

$$= 0.1027$$

Hence, the required probability is 0.8288

- iii. The total rate out of B is 1.36.

Hence the expected holding time of state B ($1/1.36$) = 0.7353 years

- iv. The probability of landing in to state D when the process moves from state B is

$$= \frac{0.04}{1.36} = 0.0294$$

- v. Let m_i denote the expected time to reach state D from state i .

As $\lambda_B = 1.36$

$$m_B = \frac{1}{1.36} + \frac{1.2}{1.36}m_A + \frac{0.12}{1.36}m_C + \frac{0.04}{1.36}m_D$$

Again $m_D = 0$ then

$$m_B = \frac{1}{1.36} + \frac{1.2}{1.36}m_A + \frac{0.12}{1.36}m_C \dots\dots\dots (i)$$

Also we know,

$$m_C = \frac{1}{0.35} = 2.857$$

Hence equation (i) becomes

$$m_B = \frac{1}{1.36} + \frac{1.2}{1.36}m_A + \frac{0.12}{1.36} \times 2.857$$

$$m_B = 0.9874 + \frac{1.2}{1.36}m_A \dots\dots\dots (ii)$$

Since $\lambda_A = 0.08$

$$m_A = \frac{1}{0.08} + \frac{0.05}{0.08}m_B + \frac{0.03}{0.08}m_D$$

Since $m_D = 0$ then

$$m_A = \frac{1}{0.08} + \frac{0.05}{0.08}m_B \dots\dots\dots (iii)$$

Equating (ii) and (iii) we get

$$m_B = 0.9874 + \frac{1.2}{1.36} \left(\frac{1}{0.08} + \frac{0.05}{0.08} m_B \right)$$

$$m_B \left(1 - \frac{1.2}{1.36} \times \frac{0.05}{0.08} \right) = 0.9874 + \frac{1.2}{1.36} \times \frac{1}{0.08}$$

$$m_B = 26.79$$

Hence, the expected time to reach state D from State B 26.79 years

[Total 13 marks]

Solution 9 :-

i. Limitations of Graphical Method of Graduation

The student has carried out graduation using a graphical approach. These rates are expected to be used as the basis for published results of the company. The key areas of risk or concern with this approach are listed below:

1. Scope for individual bias and human error

Graphical graduation requires a great deal of skill and patience and relies heavily on the judgment of the individual practitioner. Given that the rates have been derived by a student, this may pose a risk that the derived rates are inappropriate because the student may not have the necessary expertise to capture all the nuances.

Given also the subjective nature of the method, the graduated rates may suffer from any individual bias that the student may have.

2. Difficulty in justification of chosen method

Since the graduated rates are to be used as assumptions for publicly reported results, it is quite likely that the chosen methods for derivation of assumptions come under great deal of scrutiny from both internal and external reviewers. In view of this, the choice of graphical method of graduation could be particularly hard to justify – particularly since a long and credible history of mortality experience is available which should enable use of more robust approaches such as graduation using a parametric formula or by reference to a standard table.

3. Difficulty in obtaining results that are both smooth and adhere satisfactorily to the data

Due to the nature of the method, it is extremely difficult to obtain results that are both smooth and adhere to satisfactorily to the data. It is usually necessary to make several attempts, and to adjust the results by hand (rather than by re-drawing the curve) to restore smoothness (a process sometimes called hand-polishing).

4. Possibility of inappropriate shape of the curve

The shape of the curve is completely flexible when using Graphical Graduation method (since it is essentially drawn “by hand”). This may lead to a shape of the curve (or portion of the curve) that is inappropriate for representing human mortality or the underlying data. Non-intuitive results may be more difficult to explain when disclosing to the public.

5. Limited accuracy of graduated rates (up to around 3 significant figures)

Since the graduated rates have to be read from the axis of the graph, the accuracy of the rates may be limited only up to 3 significant figures. This may be severely limiting as the rates would be used to model future mortality dependent cash-flows that may require a greater degree of accuracy in the underlying probabilities. This could lead to significant mis-estimation of the financial results, particularly if the Rupee amounts underlying the cash-flows are large.

6. Subjectivity and lack of auditability in results

It is quite possible using the Graphical method that different results are obtained even if using the same underlying data and methodology if the graduation is carried out more than once or by different individuals. This leads to a great degree of subjectivity and lack of auditability in the results, which may be unacceptable particularly if the results are expected to be publicly reported.

7. Scope of mis-leading results from formal tests on the graduated results

When applying the chi-square test to determine the goodness of fit of the graduated rates, it is difficult to determine the number degrees of freedom when graduation has been done by Graphical Method. This may pose a risk that the statistical tests on graduated rates result in results that are either misleading or worse, subject to manipulation by changing the number of degrees of freedom.

ii. Advantages of Graduation by Parametric Formula and additional challenges

The alternative method proposed uses the Makeham parameterization for representing human mortality which is generally regarded as a suitable representation particularly for middle ages. This approach helps overcome several limitations in the Graphical Method listed above, such as:

1. Eliminates scope of bias

Since the proposed alternative method relies on a formal mathematical parameterization of the graduated rates from crude rates, this removes the scope for individual bias within the final results. Moreover, it is possible to extend statistical theory to undertake the necessary optimization for the proposed Makeham parameterization of the graduated rates.

This also makes the choice of method more easily justifiable than using Graphical graduation as the method can be demonstrated to be fairly robust given a suitably large volume of data.

2. Easy to achieve smoothness

Given the nature of the parameterization, since only a small number of parameters are used (only three) and the formula is expressed in terms of a simple exponential, the resulting graduated rates will automatically be smooth. Therefore, we do not have to worry about numerous iterations as required under the Graphical method to ensure reasonably smooth rates that also adhere to the underlying data.

3. Greater accuracy of graduated rates since calculated based on a mathematical formula.
4. Removes subjectivity and the method and results are demonstrably auditable – which would be desirable particularly from a public disclosure point of view. In addition, the calculations can be computerized, thus removing scope of human error and introducing potential for industrialization of the processes.
5. Satisfactory goodness of fit – since the parameters will be calibrated based on the underlying data, the goodness of fit will generally be satisfactory.
6. Use of parametric formula may result in the possibility of easier comparison across different rates (for example, if comparing rates against results for other publicly reported companies) if a similar formula is used.

Additional challenges:

1. May be time-consuming and difficult to explain methodology to the public.
2. The proposed formula may not be appropriate for the whole age range. There may be some special features within the data that need to be incorporated which would not be straight forward using a parametric approach.

iii. Tests on graduated rates

The following tests may be considered on the graduated rates derived from either method:

1. Testing smoothness of graduated rates using third differences

We could test for smoothness by looking at the third differences in the graduated rates to check whether these are small in quantity compared to the graduated rates themselves as well as progress regularly. This test may be of less importance for graduation using the parametric approach as that should ideally result in rates that are automatically smooth but would be more relevant for the graduated rates using graphical approach.

2. Chi square test

A chi square test can be used to decide whether the observed numbers of individuals who fall into specified categories are consistent with a model that predicts the expected numbers in each category. It is a test for **overall goodness of fit**.

In particular, we could test for number of lives dead/alive at each age: This would enable us to test whether the observed numbers of deaths and survivors at each age are consistent with the numbers predicted by the probabilities q_x for a particular graduation.

3. Standardised deviations test

This test may be used in conjunction with the Chi-square test to overcome its failure in detecting a number of excessively large deviations. This test can detect overall goodness of fit. Where the test reveals a problem this may be due to under/overgraduation, heterogeneity or duplicates.

4. Signs test

The signs test is a simple test for overall bias, i.e. whether the graduated rates are too high or too low.

5. Cumulative deviations

The cumulative deviations test detects overall bias or long runs of deviations of the same sign.

6. Grouping of signs test

This test can be used to test for overgraduation. The grouping of signs test (also called Stevens' test) detects "clumping" of deviations of the same sign. It relies on some simple combinatorics.

7. Serial correlations test

This test could also be used to test for overgraduation. The serial correlations test detects grouping of signs of deviations.

Additional considerations

Additional considerations should be taken into account before making the final recommendation regarding choice of assumption:

1. Consistency of final assumption with

- Current assumptions;
- Benchmark against peers (material difference in assumption, if observed should be justifiable)
- Benchmark against standard industry table

2. The results using the graduated rates are required to be publicly disclosed. Therefore, the methodology used in determining the assumption should be robust and justifiable when questioned by both internal and external reviewers. It should also be easy to explain to analysts and general public.

[Total 16 Marks]

Solution 10 :-

We are required to construct a model to determine the probability of survival after 4 years from the date of diagnosis of cancer.

We can determine this by considering the probability of deaths over the investigation period up to 4 years. The time-intervals (in months) provided are:

$0 < t \leq 1/12$: first month, spent in hospital

$1/12 < t \leq 3/12$: time interval between discharge and first check-up (two months after discharge)

$3/12 < t \leq 7/12$: time interval between first (two months after discharge) and second check-up

$7/12 < t \leq 13/12$: time interval between six months and 1 year after discharge

$13/12 < t \leq 25/12$: time interval between 1 year and 2 years after discharge

$25/12 < t \leq 61/12$: time interval between 2 and 5 years after discharge

The total number of patients diagnosed with cancer, at time, $t = 0$ are: $553 + 454 + 773 + 897 = 2,677$.

Of these, $4 + 3 + 8 + 10 = 25$ died within the first month itself, when they were still admitted in the hospital. Therefore, probability of death within the first month can be written as:

$${}_{1/12}q_0 = \frac{25}{2,677} = 0.934\%$$

Thus, the number of patients discharged at the end of first month were $2,677 - 25 = 2,652$

Of these, 8% discontinued their treatment, therefore number of lives at risk after discharge
 $= 2,652 \times (1 - 8\%) = 2,439.8$

The total number of reported deaths within two months after discharge $= 21 + 14 + 42 + 40 = 117$.

10% of deaths are not reported, therefore actual deaths within the first two months after discharge
 $= \frac{117}{(1 - 10\%)} = 130$

Consequently, we can determine the probability of death as number of deaths / number of lives at risk. i.e.

$${}_{3/12}q_{1/12} = \frac{130}{2,439.8} = 5.328\%$$

Thus, the number of lives alive and at risk at the end of three months were $2,439.8 - 130 = 2,309.8$

Of these, 5% discontinued their treatment, therefore number of lives at risk are reduced to $= 2,309.8 \times (1 - 5\%) = 2,194.3$

The total number of reported deaths within six months after discharge $= 33 + 38 + 76 + 100 = 247$.

However, of these, 117 were already reported within the first two months. Therefore the reported deaths for the time interval $\frac{3}{12} < t \leq \frac{7}{12} = 247 - 117 = 130$

Allowing for 10%, unreported deaths, total deaths for the time interval $\frac{3}{12} < t \leq \frac{7}{12} = 130 / 90\% = 144.4$

Therefore, the probability of death over the time interval $\frac{3}{12} < t \leq \frac{7}{12}$ can be written as:

$${}_{7/12}q_{3/12} = 144.4 / 2,194.3 = 6.583\%$$

We can similarly extend the above model to the remaining time-intervals to obtain the corresponding probabilities of deaths. Therefore we get:

	$0 < t < 1/12$	$1/12 < t < 3/12$	$3/12 < t < 7/12$	$7/12 < t < 13/12$	$13/12 < t < 25/12$	$25/12 < t < 61/12$
Number of patients at the start of time interval	2,677.0	2,652.0	2,309.8	2,049.9	1,784.5	1,433.3
Proportion of patients discontinuing treatment at start of time interval	0%	8%	5%	2%	0%	0%
Number of patients at risk at the start of time interval	2,677.0	2,439.8	2,194.3	2,008.9	1,784.5	1,433.3
Number of reported deaths	25.0	117.0	130.0	202.0	316.0	664.0
Proportion of deaths that are not reported	0%	10%	10%	10%	10%	10%
Total number of deaths	25.0	130.0	144.4	224.4	351.1	737.8
Probability of death over time interval	0.934%	5.328%	6.583%	11.172%	19.676%	51.472%
Number of patients at the end of time interval	2,652.0	2,309.8	2,049.9	1,784.5	1,433.3	695.6
Probability of survival to the end of time interval	100.0%	99.1%	93.8%	87.6%	77.8%	30.3%

Now, we require probability of survival for 4 years, or survival up to $t = \frac{48}{12}$.

In the final time-interval, we have the probability of death over the 36 month (or 3 year) time interval,
 $25/12 < t \leq 61/12$.

However, we only require probability of death up to the end of fourth year, i.e. up to 48th month. Using the uniform distribution of deaths assumption, we can determine the probability of death for the 23 month period leading up the end of fourth year ($48 - 25 = 23$) as:

$${}_{48/12}q_{25/12} = \frac{(48 - 25)}{36} \times {}_{61/12}q_{25/12} = \frac{23}{36} \cdot 51.472\% = 32.9\%$$

Therefore, we can finally determine the probability of survival for 4 years after diagnosis of cancer as:

$$\begin{aligned} & 1 \times (1 - {}_{1/12}q_0)(1 - {}_{3/12}q_{1/12})(1 - {}_{7/12}q_{3/12})(1 - {}_{13/12}q_{7/12})(1 - {}_{25/12}q_{13/12})(1 - {}_{48/12}q_{25/12}) \\ &= (1 - 0.934\%)(1 - 5.328\%)(1 - 6.583\%)(1 - 11.172\%)(1 - 19.676\%)(1 - 32.885\%) \\ &= 42.0\% \end{aligned}$$

[Total 14 marks]
